

Week 1 · Comprehensive Study Guide

BLOOD: Anatomy, Physiology & Clinical Applications

Virginia University of Integrative Medicine · Dr. Paymon Sadrolsadot ND PhD (TCM)

§1 Blood Overview	§2 Plasma	§3 Red Blood Cells
§4 White Blood Cells	§5 Platelets	§6 Hemostasis
§7 Disorders	§8 Clinical/TCM	§9 Quiz & Mnemonics



§1 · Blood Overview

Physical Characteristics

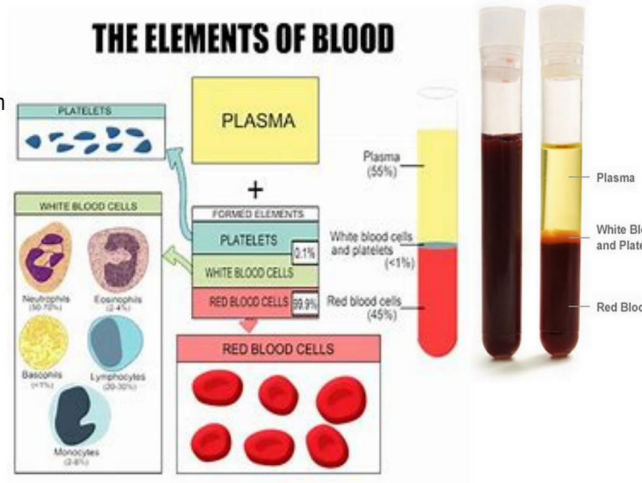
- **Appearance:** Sticky, opaque; scarlet when O₂-rich; dark red when deoxygenated
- **Taste/Smell:** Metallic and salty due to iron (hemoglobin) and electrolytes
- **Density:** More dense than water; ~5× more viscous than water
- **pH:** Slightly alkaline: 7.35 – 7.45
- **Volume:** Males: 5–6 L · Females: 4–5 L

Blood as Connective Tissue

Blood is the body's **only fluid connective tissue**. Like all connective tissues, it consists of **living cells** (formed elements) suspended in a **non-living matrix** (plasma).

Blood

- Blood is a connective tissue composed of a liquid medium called **plasma** in which solid components are suspended.
- The **Cellular Components** of blood include:
 1. Red blood cells (RBC, **erythrocytes**)
 2. White blood cells (WBC, **leukocytes**)
 3. Platelets (**thrombocytes**)



▲ Blood composition — cellular components suspended in plasma

8 Functions of Blood

Category	Function	Key Detail
Distribution	Oxygen delivery	RBCs carry O ₂ from lungs to all body cells
Distribution	Waste removal	CO ₂ → lungs; BUN/Creatinine → kidneys
Distribution	Hormone transport	Endocrine glands → target organs via blood
Regulation	Temperature	Superficial vessel dilation/constriction; heat radiation
Regulation	pH balance	Blood proteins = buffers; bicarbonate = alkaline reserve
Regulation	Fluid volume	Plasma proteins prevent fluid leakage (osmotic pressure)
Protection	Hemostasis	Platelets + plasma proteins → clot formation
Protection	Immunity	WBCs, antibodies, complement proteins fight pathogens

■ Prof. Pearls — Blood as CT

Dr. Sadrolsadot: 'Blood has a matrix (plasma) with cellular components floating inside — exactly like other connective tissues (bone, cartilage). That's why it's classified as connective tissue.'

Key analogy: plasma = ground substance; formed elements = cells/fibers



§2 · Plasma

Composition

Plasma is the **straw-colored, non-living fluid matrix** of blood (~55% of total blood volume).

Component	%	Examples / Notes
Water	90%	Primary solvent; maintains blood fluidity
Plasma Proteins	~7%	Albumin (60%), globulins, fibrinogen — mainly made in liver
Electrolytes	~1%	Na ⁺ , K ⁺ , Ca ²⁺ , Mg ²⁺ , Cl ⁻ , HCO ₃ ⁻ ; homeostasis of pH & fluid
Nutrients	<1%	Amino acids, glucose, lipids, vitamins
Respiratory Gases	trace	O ₂ and CO ₂ dissolved in plasma
Hormones	trace	Thyroid hormone, cortisol, EPO, etc.
Waste Products	trace	Urea, creatinine, BUN, uric acid → removed by kidneys

Plasma Proteins — Deep Dive

Protein	Function	Made By
Albumin (~60%)	Osmotic pressure — keeps water in vessels; prevents edema	Liver
Alpha/Beta Globulins	Transport lipids, fat-soluble vitamins, metal ions	Liver
Gamma Globulins	Antibodies (immunoglobulins) — immune defense	Plasma cells (B-cells)
Fibrinogen (~4%)	Clot formation: fibrinogen → fibrin mesh (coagulation)	Liver
Other clotting factors	Factors VIII, IX, etc. — coagulation cascade	Liver (mostly)

■ Clinical: Kwashiorkor

Severe protein malnutrition → decreased albumin → ↓ osmotic pressure → water leaves vessels and pools in abdomen (edema, 'pot belly').

The swollen belly is FLUID, not fat. Also causes brittle hair, muscle wasting.

Treatment: protein reintroduction.

§3 · Red Blood Cells (Erythrocytes)

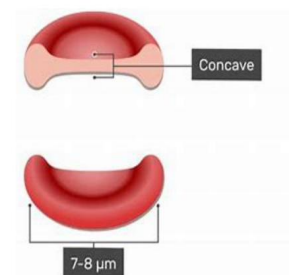
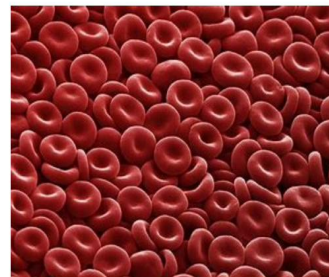
Structure

Mature RBCs are **biconcave discs** — flattened with depressed centers — optimally shaped for gas exchange and flexibility in narrow capillaries.

- **Anucleate:** No nucleus, no organelles → 97% of volume = hemoglobin
 - No nucleus means they cannot divide or repair themselves
- **Biconcave Shape:** Large surface area relative to volume
 - Every internal point is close to the membrane → rapid gas exchange
- **Flexible Membrane:** Can deform to squeeze through capillaries
 - Some capillaries are so narrow that only 1 RBC passes at a time
- **Anaerobic ATP:** Lack mitochondria; generate energy without O₂
 - They carry oxygen but do NOT use it — maximum delivery efficiency

Structural Characteristics

- **Small biconcave disc** like cells
 - Flattened discs with depressed centers
 - Lighter in color in the center
- Mature erythrocytes lack a nucleus
 - **Anucleate with no organelles**
 - **Bags of hemoglobin**
- **Able to deform** as needed to go in and out of vessels
 - Twist, turn, become cup shaped
- Erythrocytes are the **major factor contributing to blood viscosity.**



▲ RBC structure: biconcave disc, anucleate, flexible

Hemoglobin — Structure & Function

Hemoglobin (Hb) is the oxygen-carrying molecule that fills the RBC. Each Hb molecule has **4 heme groups** (each with a central **iron atom, Fe²⁺**) and **4 globin chains** (protein).

Table 17.1 Composition of Plasma (1 of 2)

Table 17.1 Composition of Plasma	
Water	90% of plasma volume; dissolving and suspending medium for solutes of blood; absorbs heat
Solutes	
Electrolytes	Most abundant solutes by number; cations include sodium, potassium, calcium, magnesium; anions include chloride, phosphate, sulfate, and bicarbonate; help to maintain plasma osmotic pressure and normal blood pH
Plasma proteins	8% (by weight) of plasma; all contribute to osmotic pressure and maintain water balance in blood and tissues; all have other functions (transport, enzymatic, etc.) as well
• Albumin	60% of plasma proteins; produced by liver; main contributor to osmotic pressure
• Globulins	36% of plasma proteins
alpha, beta	Produced by liver; most are transport proteins that bind to lipids, metal ions, and fat-soluble vitamins
gamma	Antibodies released by plasma cells during immune response
• Fibrinogen	4% of plasma proteins; produced by liver; forms fibrin threads of blood clot

▲ Hemoglobin structure: heme groups with central iron (Fe) and globin protein chains

Fact	Detail
Oxygen capacity	Each Hb carries 4 O ₂ molecules
Hb per RBC	250 million Hb molecules per single RBC
O ₂ per RBC	1 BILLION O ₂ molecules per RBC ('we are all billionaires!' – Dr. S.)
Iron role	Fe ²⁺ binds O ₂ reversibly. Iron deficiency → poor Hb → anemia
Why inside RBC?	Prevents Hb breakdown, prevents leakage, prevents ↑ viscosity
Makes blood red	Oxyhemoglobin = scarlet; Deoxyhemoglobin = dark red/maroon

Normal Reference Ranges

Measurement	Women	Men
RBC count	4.2–5.4 million/mL	4.7–6.1 million/mL
Hemoglobin (Hb)	12–16 g/100 mL	13–18 g/100 mL
Hematocrit (HCT)	~42%	~47%
Reticulocytes	1–2% of RBCs	1–2% of RBCs

Blood Viscosity & RBC Count

RBC number is the **primary determinant** of blood viscosity (thickness):

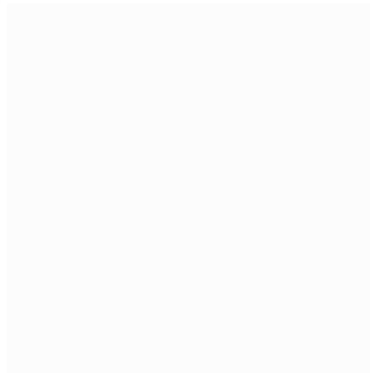
- ↑ **RBCs** → ↑ viscosity → slower flow → **risk of blood clots**
- ↓ **RBCs** (anemia) → ↓ viscosity → faster flow → poor O₂ delivery

■ Clinical: Blood Thinners

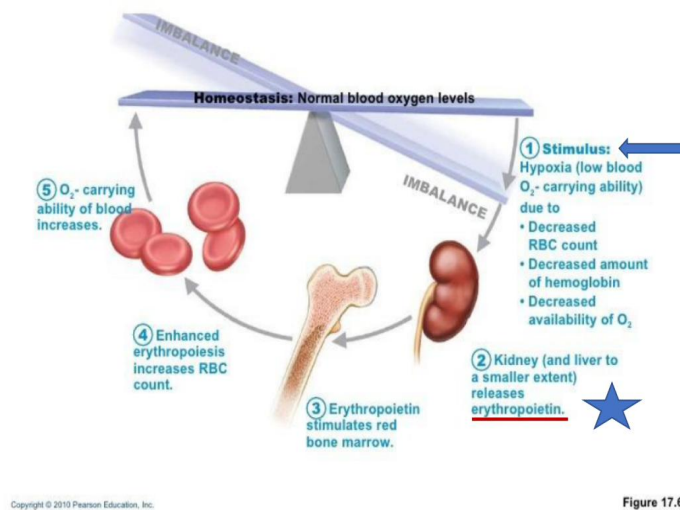
Blood thinners (aspirin 81mg, heparin, warfarin) reduce platelet aggregation and clotting cascade risk.

They do NOT destroy RBCs or reduce their number.

Prescribed for: DVT prevention, post-MI, atrial fibrillation, high cardiovascular risk.



Erythropoiesis — RBC Production



▲ *Hematopoiesis: all blood cells arise from multipotential hematopoietic stem cell (hemocytoblast)*

1. **Stem Cell:** Multipotential hematopoietic stem cell (Hemocytoblast) in RED bone marrow
2. **Myeloid Stem Cell:** Committed to myeloid pathway (cannot revert)
3. **Early Erythroblast:** Large nucleus; begins synthesizing hemoglobin
4. **Nucleus Shrinks:** Hemoglobin accumulates; nucleus progressively ejects
5. **Reticulocyte:** Immature RBC; nucleus ejected; still has ribosomes; released to blood
6. **Mature Erythrocyte:** Ribosomes degraded within 2 days in blood; fully mature biconcave disc

Key Production Facts

- Total process (stem cell → reticulocyte): ~15 days
- Reticulocyte → mature RBC in blood: ~2 days
- Reticulocytes = 1–2% of all RBCs (healthy baseline)
- Bone marrow produces ~100 billion new cells / day (~1 ounce of blood)
- RBC lifespan: 100–120 days → cleaned by SPLEEN (+ liver to lesser extent)
- REGULATION hormone: Erythropoietin (EPO) — mainly from kidneys, stimulated by hypoxia
- Dietary requirements: Iron (heme), Amino acids (globin), B12 + Folate (DNA synthesis), Vitamin C

■ Clinical: Reticulocyte Count

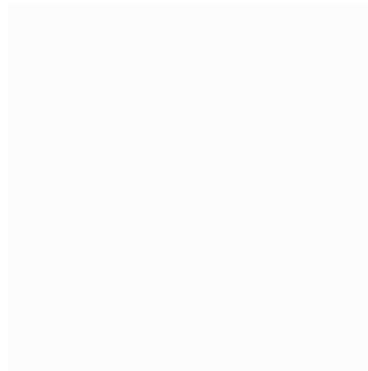
Normal reticulocyte count: 1–2% of RBCs.

Elevated (e.g. 10%) → bone marrow is overproducing, suggesting: hemolytic anemia (destroying fast), blood loss (compensating), iron-replenishment after deficiency.

Low reticulocyte count → bone marrow underproduction (aplastic anemia, B12 deficiency).

Dietary Requirements for Erythropoiesis

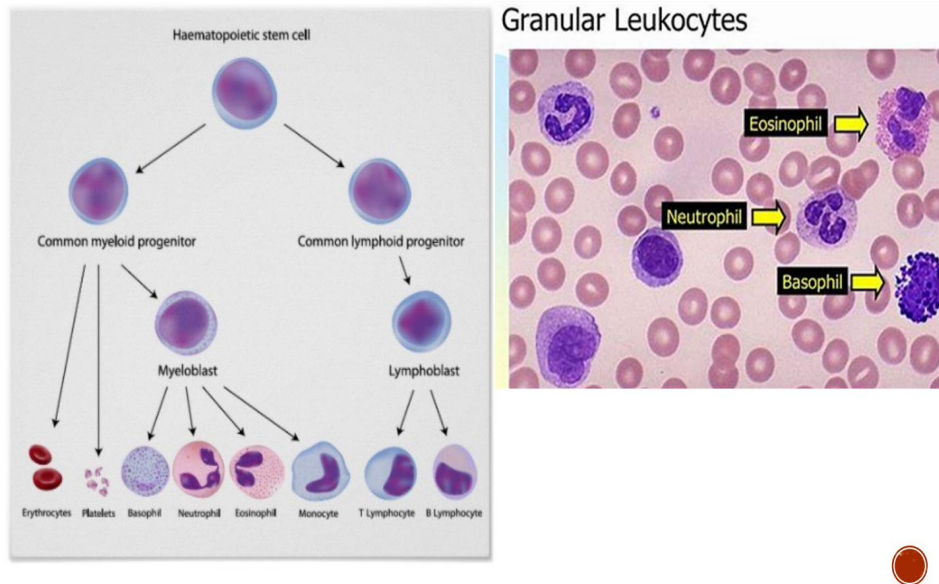
Nutrient	Role	Deficiency Consequence
Iron (Fe ²⁺)	Center of heme group; binds O ₂	Iron Deficiency Anemia (IDA) — most common
Amino Acids	Build globin protein chains	Poor hemoglobin synthesis; protein malnutrition
Vitamin B12	DNA synthesis for cell division	Pernicious anemia; megaloblastic anemia
Folate (B9)	DNA synthesis (esp. during cell division)	Megaloblastic anemia; neural tube defects (pregnancy)
Vitamin C	Enhances iron absorption in GI tract	Reduced iron absorption → secondary anemia
Erythropoietin	Hormone signaling bone marrow to produce RBCs	Renal anemia (kidneys fail to produce EPO)



§4 · White Blood Cells (Leukocytes)

Overview

Leukocytes are the **only TRUE cells** in blood — they have a nucleus and all organelles. They make up **<1% of blood volume** but are the body's mobile immune army.



▲ WBC types: *granulocytes (neutrophils, eosinophils, basophils) and agranulocytes (lymphocytes, monocytes)*

Never Let Monkeys Eat Bananas

Neutrophils · Lymphocytes · Monocytes · Eosinophils · Basophils (high → low %)

WBC Classification Table

WBC Type	%	Category	Function
Neutrophils	50–70%	Granulocyte	Bacteria slayers; most numerous WBC
Lymphocytes	25–33%	Agranulocyte	T-cells (virus-infected cells); B-cells (antibodies)
Monocytes	3–8%	Agranulocyte	Become macrophages; fight viruses & chronic infections
Eosinophils	2–4%	Granulocyte	Parasitic worms, allergies, asthma
Basophils	0–0.75%	Granulocyte	Release histamine (inflammatory response)

WBC Mobilization

- Follow chemical trails from damaged tissue (**chemotaxis**)
- Exit blood vessels via **diapedesis** ('leaping across' the vessel wall)
- **Leukocytosis**: ↑ WBC count (usually infection); can double in 1 hour
- **Leukopenia**: ↓ WBC count (often drug-induced, e.g. chemotherapy)

Detailed WBC Functions

Neutrophils (50–70%)

First responders; phagocytize bacteria; release antimicrobial chemicals; multi-lobed nucleus; shortest lifespan (hours–days)

Lymphocytes (25–33%)

T-cells: directly kill virus-infected cells & cancer cells. B-cells: produce antibodies (immunoglobulins). NK cells: natural killers.

Monocytes (3–8%)

Largest WBC; become MACROPHAGES in tissues; engulf large pathogens, dead cells, cellular debris; important in chronic infections

Eosinophils (2–4%)

Combat parasitic worm infections; modulate allergic response; release toxic proteins against parasites; elevated in asthma/allergies

Basophils (0–0.75%)

Rarest WBC; release HISTAMINE (vasodilation, inflammatory response); also release heparin (anti-coagulant); similar to mast cells

§5 · Platelets (Thrombocytes)

Structure & Formation

- **Not true cells** — cytoplasmic fragments of large cells called **Megakaryocytes**
- **Anucleate** → age rapidly; degenerate in ~8–10 days if unused
- Contain granules with chemicals essential to clotting
- Normally kept **inactive** by endothelial-secreted **Nitric Oxide (NO)**
- Count: 150,000–500,000 per μL of blood
- Regulated by hormone: **Thrombopoietin**
- Formation: Megakaryocytes press against sinusoid capillaries in red bone marrow → send cytoplasmic extensions into blood → extensions rupture → platelets released

§6 · Hemostasis

Hemostasis = stopping blood loss after vessel injury. Must be **fast, localized, and controlled**. Three sequential steps:

Table 17.2 Summary of Formed Elements of the Blood (2 of 2)

Table 17.2 Summary of Formed Elements of the Blood (continued)					
CELL TYPE	ILLUSTRATION	DESCRIPTION*	CELLS/ μL (mm^3) OF BLOOD	DURATION OF DEVELOPMENT (D) AND LIFE SPAN (LS)	FUNCTION
Agranulocytes					
• Lymphocyte		Spherical or indented nucleus; pale blue cytoplasm; diameter 5–17 μm	1500–3000	D: days to weeks LS: hours to years	Mount immune response by direct cell attack or via antibodies
• Monocyte		U- or kidney-shaped nucleus; gray-blue cytoplasm; diameter 14–24 μm	100–700	D: 2–3 days LS: months	Phagocytosis; develop into macrophages in the tissues
Platelets		Discoid cytoplasmic fragments containing granules; stain deep purple; diameter 2–4 μm	150,000–400,000	D: 4–5 days LS: 5–10 days	Seal small tears in blood vessels; instrumental in blood clotting

*Appearance when stained with Wright's stain.

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▲ Hemostasis: 3 steps — vascular spasm, platelet plug, coagulation (fibrin mesh)

Step 1 · Vascular Spasm

Immediate constriction of the injured blood vessel → **reduces blood flow** to injury site.

Triggers:

- Direct injury to vascular smooth muscle
- Chemicals released by endothelial cells and platelets (serotonin, thromboxane)
- Reflexes initiated by local pain receptors

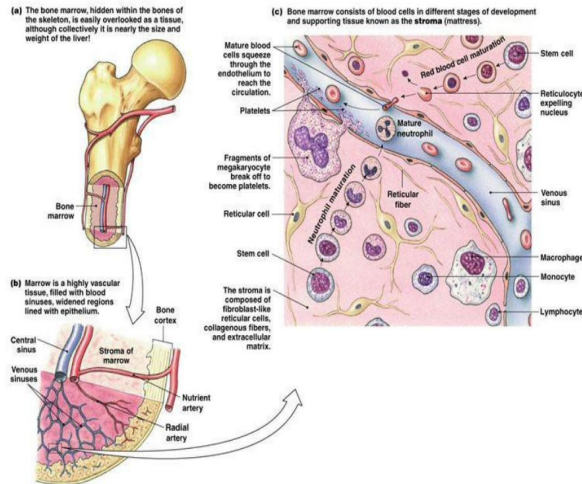
Step 2 · Platelet Plug Formation (within ~1 minute)

Exposed **collagen fibers** (from damaged vessel wall) attract platelets.

- Platelets adhere to collagen → swell → form spike-like processes
- Release **ADP** → calls more platelets to site
- Release **serotonin + thromboxane** → enhance vascular spasm
- Aggregate into a loose but effective platelet plug

Step 3 · Coagulation (Fibrin Mesh Formation)

Focus on ... Bone Marrow



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Figure 16-4a

▲ *Coagulation cascade: clotting factors activate in sequence → fibrin mesh seals wound*

A cascade of clotting factors (plasma proteins, mostly from liver) activates in sequence. The final result:

- **Prothrombin** (inactive) → **Thrombin** (active enzyme)
- **Fibrinogen** (soluble plasma protein) → **Fibrin** (insoluble mesh)
- Fibrin mesh traps RBCs and platelets → stable, sealed blood clot

■ Acupuncture Clinical — Post-Needle Bleeding

When an acupuncture needle severs a small vessel, the natural hemostasis sequence occurs.

PRESS with cotton ball — do NOT massage.

Reason: The fibrin mesh and platelet plug form within ~30 seconds to 1 minute but remain fragile. Massage disrupts the forming clot → re-bleeding → bruising the next day.

Rule: Firm steady pressure for 30 seconds; release when bleeding stops.

§7 · Disorders of Blood

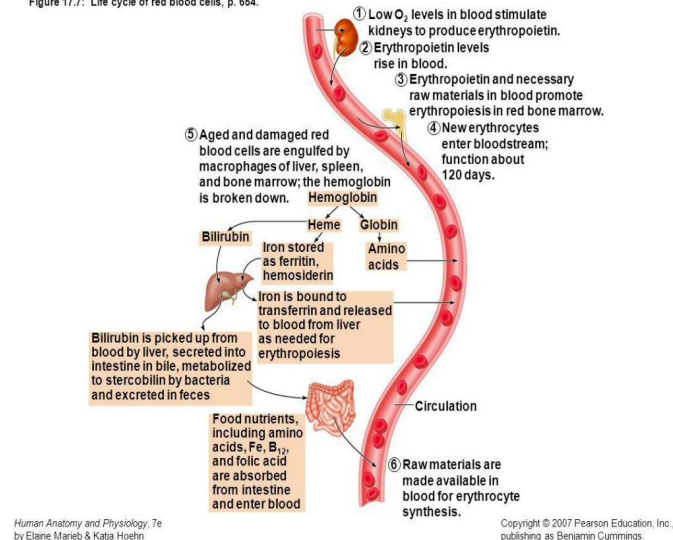
Erythrocyte Disorders

Anemias

Anemia = reduced oxygen-carrying capacity. A **sign** of underlying pathology, not a disease itself. Symptoms: fatigue, pallor, dyspnea, palpitations, cold extremities.

Type	Cause	Notes
Iron Deficiency (IDA)	Insufficient iron → poor hemoglobin	Most common; fatigue, pallor, brittle nails
Pernicious / B12 Def.	Lack of Intrinsic Factor (IF) → B12 malabsorption	Large, fragile RBCs; neurological symptoms
Aplastic Anemia	Bone marrow failure → no stem cells	Pancytopenia; very serious
Sickle Cell Anemia	Genetic HbS mutation → sickle-shaped RBCs	Vaso-occlusive crises, hemolysis
Thalassemia	Genetic defect in globin chain production	Mediterranean, African, SE Asian populations
Hemolytic Anemia	Premature RBC destruction	Jaundice, splenomegaly
Renal Anemia	Low erythropoietin from kidney disease	EPO supplementation used in treatment

Figure 17.7: Life cycle of red blood cells, p. 654.



▲ Comparison of normal RBC vs sickle cell, and other RBC morphologies in different anemias

Polycythemia

- **Definition:** Abnormal excess of erythrocytes
- **Polycythemia Vera:** Bone marrow cancer → uncontrolled RBC overproduction
- **Secondary Polycythemia:** Normal physiological response to high altitude ($\downarrow O_2 \rightarrow \uparrow EPO$)
- **Blood Doping:** Athletes use exogenous EPO to artificially increase RBC count

- **Risk:** ↑ blood viscosity → sluggish flow → ↑ risk of thrombus, stroke, MI

Hemostatic Disorders

Disorder	Type	Mechanism	Clinical Notes
DVT (Deep Vein Thrombosis)	Thrombus	Pooling blood (immobility) → platelet aggregate in intact vessel	Long flights, bedrest; walk to prevent!
Pulmonary Embolism	Embolus	DVT clot dislodges → travels to pulmonary vessels	Life-threatening; sudden dyspnea
Stroke (CVA)	Embolus	Embolus blocks cerebral vessel	FAST: Face/Arm/Speech/Time
Thrombocytopenia	Bleeding disorder	↓ platelet count (autoimmune, drug-induced, etc.)	Petechiae, easy bruising, prolonged bleeding
Hemophilia A/B	Bleeding disorder	Factor VIII (A) or IX (B) deficiency → coagulation cascade fails	X-linked recessive; treat with factor replacement
Liver Disease	Bleeding disorder	Liver fails to produce clotting factors (fibrinogen, prothrombin, etc.)	Cirrhosis, end-stage liver disease

Anticoagulant Drugs

Drug	Mechanism	Route/Use
Aspirin (81–325 mg)	Blocks platelet aggregation (inhibits thromboxane A ₂)	Oral; daily prophylaxis in cardiovascular risk
Heparin	Enhances Antithrombin III → inhibits thrombin & factor Xa	Injectable (IV/SQ); hospital setting, acute DVT
Warfarin (Coumadin)	Interferes with Vitamin K → ↓ synthesis of clotting factors II, VII, IX, X	Oral; long-term anticoagulation post-discharge

■ Vitamin K Note

Warfarin blocks Vitamin K → reduces production of clotting factors.

Newborns receive Vitamin K injection at birth because: their gut has no bacteria yet to produce K₂, and without it, any bleeding could be fatal.

TCM/Dietary note: high Vitamin K foods (leafy greens, natto) can interact with warfarin.

§8 · Clinical & TCM Connections

Integrating Western hematology with Traditional Chinese Medicine perspectives:

Blood (Xue 血) in TCM

Blood is a dense, nourishing Yin substance. It moistens tissues, nourishes the mind (Shen), and circulates through vessels guided by Heart Qi. Blood deficiency (Xue Xu) mirrors Western anemia in many symptoms: pallor, fatigue, poor memory, dizziness, palpitations.

Iron & Spleen Qi

The Spleen in TCM transforms food essence into Blood. Iron deficiency anemia often corresponds to Spleen Qi Deficiency: poor digestion, low energy, loose stools. Treatment includes tonifying Spleen Qi (SP-6, ST-36, BL-20) alongside dietary iron support.

EPO & Kidney Yang

Erythropoietin is made primarily by the kidneys. TCM recognizes Kidney as the source of marrow and Essence (Jing) — directly paralleling bone marrow's role in hematopoiesis. Renal anemia corresponds to Kidney deficiency.

Hemostasis & Blood Stasis (Yu Xue 瘀血)

When hemostasis is dysregulated — excess clotting — TCM interprets this as Blood Stasis. Acupuncture points like SP-10 (Xuehai — 'Sea of Blood'), BL-17, and LV-3 are used to invigorate Blood and resolve stasis.

DVT Prevention

Long immobility → Blood Stasis in channels. TCM recommends movement, leg elevation, and Blood-invigorating herbs (e.g., Dan Shen, Hong Hua) for prevention and recovery.

Acupuncture Needling & Hemostasis

Every needle insertion risks microvessel injury. Understanding the three-step hemostasis process explains why we apply firm, steady pressure (not massage) post-needling — protecting the fragile platelet plug and forming fibrin mesh.

§9 · Quiz Review & Mnemonics

Key Mnemonics

NEVER LET MONKEYS EAT BANANAS

WBC order high→low: Neutrophils, Lymphocytes, Monocytes, Eosinophils, Basophils

3-2-2 FUNCTIONS

Blood functions: 3 Distributing (O₂, waste, hormones) · 3 Regulating (temp, pH, volume) · 2 Protecting (hemostasis, immunity)

HEME = IRON + O₂

Heme group: iron (Fe²⁺) at center binds 4 O₂ molecules. 1 Hb = 4 heme. 1 RBC = 250M Hb = 1 BILLION O₂

RED=LIVE, YELLOW=FAT

Red bone marrow = hematopoiesis. Yellow bone marrow = fat storage. Red converts to yellow with age.

V-P-C (Vascular-Plug-Clot)

Hemostasis sequence: Vascular spasm → Platelet plug → Coagulation (fibrin mesh)

BLAST = YOUNG

Any term ending in '-blast' means young/immature cell (hemocytoblast = young blood stem cell)

Practice Quiz — Lecture Style

◆ Check Your Understanding

1. List 2 protective functions of blood.
2. Why is blood classified as connective tissue?
3. Are plasma proteins used as fuel by cells? Explain.
4. What are the major components of blood?
5. Describe the process of erythropoiesis step by step.

◆ Check Your Understanding

1. How many O₂ molecules can one RBC carry? Show the math.
2. What part of hemoglobin binds to oxygen?
3. What hormone regulates RBC production? Where is it made?
4. Explain the lifecycle of an erythrocyte from stem cell to destruction.
5. What does a reticulocyte count of 10% suggest clinically?

◆ Check Your Understanding

1. Which WBC turns into macrophages in tissue?

2. Which WBC mobilizes in response to parasite infection?
3. What are the 2 categories of WBCs? Name all 5 types.
4. How do platelets differ from other formed elements?
5. What hormone regulates platelet formation?

◆ Check Your Understanding

1. Describe the 3 steps of hemostasis.
2. What is the difference between fibrinogen and fibrin?
3. What is the difference between thrombus and embolus?
4. Define DVT. What causes it? What are the risks if it dislodges?
5. How does warfarin work? How does heparin work? How does aspirin work?
6. Why do we apply steady pressure (not massage) after removing an acupuncture needle?

Additional Study Resources

- Tortora & Derrickson — Principles of Anatomy & Physiology, Ch. 17 (Blood)
- Khan Academy: Blood — hematocrit, red blood cells, white blood cells, platelets
- Osmosis: Anemias (high-yield videos for each anemia type)
- AHA Guidelines on Anticoagulation (for clinical context)
- Deadman, Al-Khafaji — A Manual of Acupuncture (SP-10, BL-17 for Blood Stasis)
- Huang Di Nei Jing — Chapter on Blood and its relationship to Heart and Spleen
- AMBOSS: Hemostasis, Coagulation Cascade (free clinical study)
- Anki Deck: 'Anatomy & Physiology Blood' (available on AnkiWeb)

■■■■■ · *'Blood flows without ceasing' — Huang Di Nei Jing*